Every Recovery Metric Has an Invariance Class: Manufactured Thresholds and Hidden Collapses in Latent-Variable Identifiability

Anonymous authors
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Abstract

A recovery metric is defined by what it cannot see. We argue, and demonstrate twice, that every estimand of latent recovery has an *invariance class*, and that a reported result is an artifact whenever the failure mode under test lies inside that class. First, k-nearest-neighbour recovery scores depend on the embedding only through its neighbour sets, so they are bit-identically invariant to the similarity group, which contains the orthogonal gauge freedom of an isotropic prior. A separate failure of the same metric manufactures a clean "critical support overlap" $\rho^* \approx 0.37$: we derive the curve in closed form, $R^2 = 1 - 4(1-\rho)^3$, so $\rho^* = 1 - 4^{-1/3}$ is the zero-crossing of the metric's own extrapolation geometry, and we reproduce it on a *perfectly identified oracle embedding.* Second, and more instructively, the rotation-sensitive estimand we built to repair the first problem is not an independent quantity at all: it equals CCA−MCC identically, its zero set is therefore the axis-factorised maps, and it missed an entire real effect. Its blind region is moreover defined relative to a chosen ground-truth basis, so an invariance class is a property of the metric *plus a coordinate choice*, one of which is unobservable on real data. We then turn the same quantity into an instrument. With the subspace recovered, CCA−MCC $= 1 - \cos\varphi$ reads out the angle between the frame a run converged to and the chosen basis, and "no rotational preference" has the closed-form null $\varphi \sim U[0°, 45°]$, giving $E = 0.0997$ and SD $= 0.0880$. Every floor we measure sits on that law and one-sample KS rejects it nowhere, including for conditional priors that satisfy the iVAE variability condition with a well-conditioned natural-parameter matrix. Sweeping $\sigma_{\min}(L)$ directly at fixed environment count makes the floor decline monotonically, identifying $\sigma_{\min}(L)$ rather than the environment count as the axis to study, but bounds the effect at $\approx 21\%$ of the full rotation signature. Finally, in a generative process where support overlap and removable batch magnitude are provably decoupled, whether overlap governs recovery turns out to be a property of the *objective*: a faithful per-sample encoder is overlap-invariant (pre-registered, equivalence-tested, 25 seeds), while alignment objectives collapse recovery as overlap falls.

1 Introduction

Work on disentanglement and latent recovery reports numbers: an MCC, a DCI score, a k-NN regression R^2, sometimes a threshold at which recovery "breaks". Each of those numbers is a function of an embedding, and each is *invariant* to some group of transformations of that embedding. That invariance class is rarely stated, and it decides what the number can possibly show.

Our claim is simple and, we think, under-appreciated:

> Every estimand of latent recovery has an invariance class. A reported result is an artifact whenever the failure mode under test lies inside that class.

This is not a claim that particular metrics are bad. It is a claim about matching an estimand to a failure mode, and about stating the class explicitly so that a reader can check the match. We support it with two demonstrations. The second is the one we would ask a sceptical reader to weigh, because it concerns a metric we designed ourselves, argued for on correctness grounds, and gated before use.

Contributions.

1. **A measured taxonomy** (§3) of six estimands against corruptions that isolate one failure mode each. The blindness is complementary: no single estimand covers the space.
2. **A manufactured threshold, derived** (§4). k-NN transfer gives $R^2 = 1 - 4(1-\rho)^3$ with zero-crossing $\rho^* = 1 - 4^{-1/3}$, reproduced on an oracle embedding. We give the recipe that generates the constant and show it is not universal.
3. **A hidden collapse in our own estimand** (§5). The matched-information oracle gap equals CCA $-$ MCC identically; its zero set is the axis-factorised maps, and it reports no effect on a real alignment-induced collapse.
4. **Basis-dependence** (§6). The blind region is defined relative to a chosen ground-truth basis, so an invariance class is a property of the metric *plus a coordinate choice*.
5. **The floor as an instrument** (§7). CCA $-$ MCC reads out a frame angle; we give the closed-form random-frame null and use it to test the iVAE variability condition directly.
6. **An application** (§8) in which overlap and removable batch magnitude are provably decoupled, showing that whether overlap governs recovery is set by the objective.

All experiments are synthetic and fully controlled by design: the object of study is what the metrics can and cannot see, which requires knowing the ground truth exactly.

2 Setup

Write $t \in \mathbb{R}^n$ for the true latent and z for a learned embedding. We use four estimands. **k-NN-R^2** is the cross-validated R^2 predicting a true factor from z by k-nearest-neighbour regression, and **k-NN transfer** is its cross-domain version, fit on one domain and scored on the other. **CCA** is the mean canonical correlation between t and z, the weak notion of recovery up to an invertible linear map. **MCC** is the mean absolute correlation under the optimal one-to-one (Hungarian) matching of recovered to true coordinates, in Pearson and Spearman variants; the assignment is solved on a train split and scored held out. **DCI-D** is the disentanglement component of Eastwood & Williams (2018). Unless stated otherwise $n = 2$ and $N = 4000$.

3 A taxonomy of what estimands cannot see

We corrupt the *true* latent in exactly one way at a time and score every estimand. There is no training and no generative model here: these are properties of the estimands themselves. A cell at its identity value means the estimand is blind to that corruption (Figure 1).

The classes, stated. k-NN-R^2 is exactly invariant to any map preserving the k-NN *sets*, which includes the similarity group $\mathrm{Sim}(d) = \text{translations} \rtimes (O(d) \times \mathbb{R}_+)$; preservation of the ranking of pairwise distances is sufficient but not necessary, and exactness requires uniform weights and the L^2 metric. CCA is invariant to any affine map whose linear part is injective on the latent, which is strictly larger than $GL(d)$, and is therefore blind to *linear* entanglement only: information-preserving nonlinear axis mixing is visible to it.

Two scope statements. First, the three corruption classes we instantiate are *illustrative, not a partition.* Each row is an existence proof that an estimand at its identity value could not have detected that failure, not a coverage map. Second, $O(d)$ is the linear part of the stabiliser of an isotropic Gaussian prior but not its whole non-identifiability class, and the two are non-nested: the measure-preserving swirl $\theta \mapsto \theta + c\|t\|^2$ leaves $\mathcal{N}(0, I_2)$ exactly invariant, yet k-NN-R^2 plainly sees it (0.94, 0.79, 0.53 for $c = 0.3, 0.9, 2.0$), while k-NN's class contains translations and rescalings that are not prior symmetries.

Each estimand is blind to a different failure (red = within 0.05 of its identity value on a genuine failure)

	kNN-R^2	CCA	MCC-P	MCC-S	DCI-D	CCA−MCC	
identity	0.996	1.000	1.000	1.000	0.997	-0.000	—
permute axes	0.996	1.000	1.000	1.000	0.998	-0.000	*nuisance*
per-axis rescale	0.999	1.000	1.000	1.000	0.997	-0.000	*nuisance*
rotate 45°	**0.996**	**1.000**	0.707	0.685	0.000	0.293	*entanglement*
shear	**0.996**	**1.000**	0.887	0.878	0.508	0.113	*entanglement*
monotone nonlinear	**0.987**	0.885	0.885	**1.000**	**0.997**	**-0.000**	*reparametrisation*
isotropic noise	0.725	0.857	0.857	0.846	0.838	**0.000**	*info loss (axis-f.)*
fold axis	-0.062	0.531	0.531	0.507	0.594	**0.000**	*info loss (axis-f.)*
collapse axis	-0.070	0.502	0.500	0.502	0.801	**0.001**	*info loss (axis-f.)*
common-mode noise	0.537	0.658	0.442	0.428	0.038	0.216	*info loss (mixed)*
collapse rotated ax.	0.482	0.500	0.358	0.356	0.001	0.142	*info loss (mixed)*

nuisances

Figure 1: Estimand × failure mode. Each row applies one corruption to the true latent; each column is one estimand. Red marks cells within 0.05 of the estimand's identity value on a genuine failure, that is, blindness. k-NN-R^2 and CCA cannot see a rotation; CCA−MCC cannot see an axis-factorised map; MCC-Spearman and DCI-D cannot see a monotone reparametrisation.

4 Demonstration 1: a threshold that is a property of the metric

Consider a one-dimensional uniform shift: domain A has $t \sim U[0, w]$ and domain B has $t \sim U[w(1-\rho), w(2-\rho)]$, so the overlap coefficient is exactly ρ. A k-NN map fit on A and scored on B must extrapolate for the fraction $(1-\rho)$ of B lying beyond A's support, where its prediction saturates at the boundary. Integrating the resulting error,

$$E[\mathrm{err}^2] = \frac{w^2(1-\rho)^3}{3}, \qquad \mathrm{Var}(t) = \frac{w^2}{12}, \qquad R^2(\rho) = 1 - 4(1-\rho)^3,$$

so the apparent threshold is $\rho^* = 1 - 4^{-1/3} \approx 0.370$, and the width w cancels. Crucially this appears for an **oracle embedding** $z := t$, which is perfectly identified by construction: a critical threshold that a perfectly identified embedding also exhibits cannot be about identifiability (Figure 2a; mean absolute error 0.016 at $N = 8000$, measured crossing 0.377, the residual offset a finite-k smoothing bias that shrinks to 0.372 as $k/n \to 0$).

The constant is a recipe, not a universal. It factors as $4 = (1/\mathrm{Var}_{\mathrm{shape}}) \times$ (second moment of the extrapolation-error law) $= 12 \times \frac{1}{3}$ for a unit uniform, so it is set by the marginal shape and the transfer construction. A triangular marginal gives a different but stable constant; a Gaussian marginal, having no support boundary, has no stable threshold at all. Embedding geometry moves it too, and the driver is per-axis *scale* rather than dimension count (Figure 2b,c). The practical check is therefore to compute the constant for one's own marginal, dimension, scale and k before reading a threshold as a phenomenon.

Two independent failures. Rotation-blindness and this extrapolation artifact are distinct, and conflating them invites the wrong repair. A rotation-sensitive metric would *not* have avoided the threshold: it appears on an embedding with no entanglement to be sensitive to, and any estimand built from a non-extrapolating predictor scored by a variance-normalised R^2 inherits the same curve and the same constant.

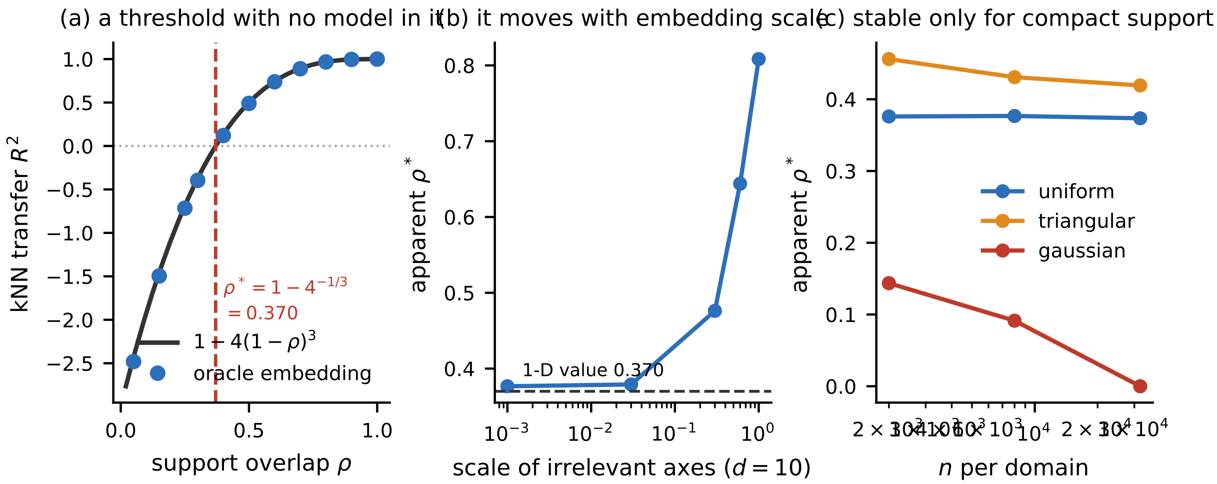

Figure 2: (a) The closed form matches an oracle embedding, so the threshold contains no model. (b) It is not scale-free in the embedding: at $d = 10$, growing the irrelevant axes drags ρ^* from 0.377 to 0.808. (c) It is a stable constant only for compact-support marginals; a Gaussian marginal's crossing drifts toward 0 as n grows.

5 Demonstration 2: a hidden collapse in an estimand we built

To repair Demonstration 1 we introduced a rotation-sensitive *matched-information oracle gap*: add isotropic noise to t until the resulting oracle carries the same information (CCA) as z, then difference MCC, so that the residual is attributable to entanglement rather than to a noisier embedding. It passed a correctness argument and a gate. It nonetheless reported no effect on a real alignment-induced collapse in which CCA fell from 0.99 to 0.70 (§8).

The reason is structural, and it reduces the estimand to something already in hand. The oracle family is isotropic noise on the *true* latent, hence axis-factorised, hence every oracle satisfies MCC = CCA. Matching information forces MCC(oracle) = CCA(z), so

$$\text{gap} \;=\; \text{CCA}(z) - \text{MCC}(z) \quad \text{identically}$$

(maximum deviation 1×10^{-4} across every corruption in Figure 1). Its exact zero set is $\{\text{MCC} = \text{CCA}\}$, the **axis-factorised** maps, in which each recovered coordinate is a function of a single true coordinate. We therefore report CCA − MCC directly and demote the oracle construction to a validation, which removes a step a reader would otherwise have to audit and makes the zero set self-evident rather than empirical.

A correction. We first characterised this blindness as "blind to information loss". That is false, and Figure 1 contains the counterexamples: our original information-loss rows merely happened to be axis-aligned, whereas common-mode noise scores 0.216 and collapsing a *rotated* axis scores 0.142, the former comparable to the 0.293 that a genuine 45° rotation produces. The correct characterisation is axis-factorisation. The alignment collapse evaded the gap because that particular failure is approximately axis-factorised in the frame the model converged to, which is a fact about the experiment rather than a property of the estimand.

6 Invariance classes depend on a coordinate choice

{MCC = CCA} is the axis-factorised set *relative to the ground-truth basis one chose.* CCA is basis-free; Hungarian matching is not, because it asks whether recovered coordinate i tracks true coordinate j. The gap therefore inherits basis-dependence entirely through MCC, and rotating the ground-truth basis moves the blind region. In synthetic work that basis is a modelling choice; on real data no privileged basis for "the true

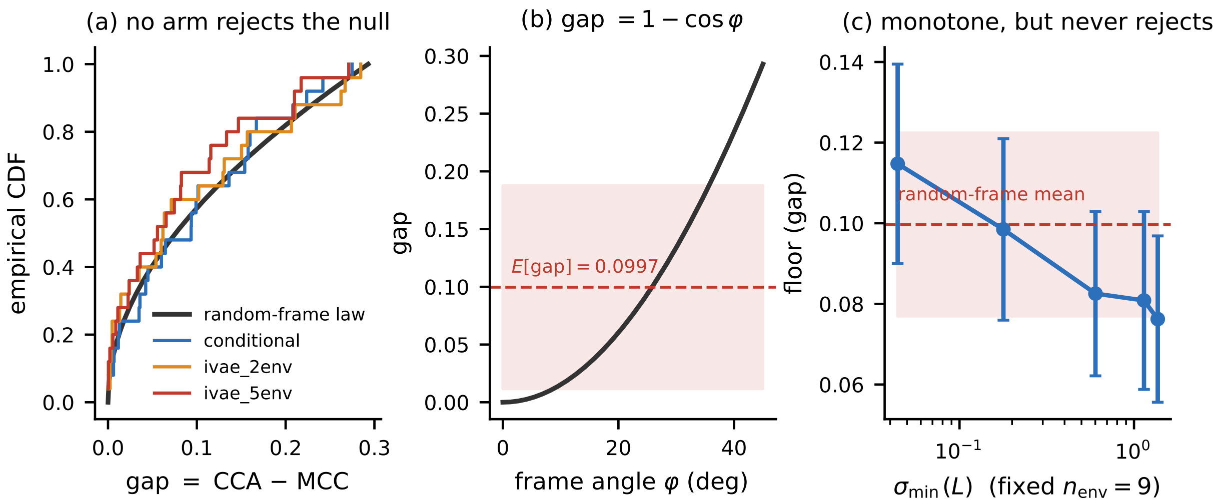

Figure 3: (a) Empirical CDFs of CCA − MCC at full overlap against the analytic random-frame law; one-sample KS rejects for no arm. (b) The gap-to-angle map. (c) Sweeping $\sigma_{\min}(L)$ at fixed $n_{\text{env}} = 9$ with total prior variability held constant: the floor declines monotonically but never leaves the law.

latent" exists, so the blind region is positioned by a choice with no observable counterpart. An invariance class is thus a property of the metric *plus a coordinate choice*, and one of those is often arbitrary.

We tested the attack this implies. In our generative process the shifted direction *is* a coordinate axis, so an alignment objective folding it is axis-factorised by construction, which would make §5 an artifact of that coincidence. Placing the shifted direction at 45° to the ground-truth basis and rerunning (128 runs, 8 seeds, with the faithful arm at the same angle as a basis-mismatch control), the gap does *not* fire for the alignment arm: its excess over the control is ≤ 0 at every overlap. The attack is closed. What surfaced instead is a larger limitation. On the faithful arm at $\rho = 1$, where the true gap is unambiguously zero, single runs span $[0.000, 0.191]$ at $\theta = 0°$ and $[0.009, 0.290]$ at $\theta = 45°$, the upper end matching what a genuine 45° rotation produces. Scoring each run against whichever candidate basis it actually converged to collapses the $\theta = 45°$ gap from 0.112 to 0.024.

7 The floor is an instrument

That spread is not estimator noise. With the subspace recovered (CCA ≈ 1), MCC registers only the angle φ between the frame a run converged to and the chosen basis; for $n = 2$, MCC $= \cos\varphi$, with Hungarian matching folding φ into $[0°, 45°]$ because a 90° rotation is a permutation. Hence

$$\text{gap} = 1 - \cos\varphi \qquad \Longleftrightarrow \qquad \varphi = \arccos(1 - \text{gap}),$$

mapping $[0, 0.293]$ onto $[0°, 45°]$ (Figure 3b). This is the rotational non-identifiability of an isotropic prior (Locatello et al., 2019) measured directly, so the quantity reports *how much axis-level identifiability a model class attains.*

A closed-form null. If the optimiser has no rotational preference then $\varphi \sim U[0°, 45°]$ and

$$F(x) = \tfrac{4}{\pi}\arccos(1 - x), \qquad E[\text{gap}] = 1 - \tfrac{2\sqrt{2}}{\pi} = 0.0997, \qquad \text{SD} = 0.0880.$$

Testing the iVAE variability condition. Khemakhem et al. (2020) require the $nk \times nk$ matrix L of natural-parameter differences to be invertible, which needs $nk + 1$ distinct auxiliary values. Across an isotropic-prior model and conditional priors with $n_{\text{env}} \in \{2, 3, 5, 9\}$, one-sample KS against the law rejects nowhere (p from 0.08 to 0.96; Figure 3a). The conclusion is stronger than "the condition does not bite as a threshold": satisfying it leaves the recovered frame *statistically indistinguishable from a uniformly random one.*

Two refinements protect that statement. First, nominal satisfaction is not effective satisfaction: at $n_{\mathrm{env}} = 5$ the matrix L is invertible but near-singular ($\sigma_{\min}(L) = 0.027$, $\kappa = 451$), whereas $n_{\mathrm{env}} = 9$ is well conditioned ($\sigma_{\min}(L) = 1.65$, $\kappa = 9.3$) and still does not reject. Second, we swept $\sigma_{\min}(L)$ *directly* at fixed $n_{\mathrm{env}} = 9$, constructing the spectrum of L with total prior variability held constant (75 runs, 15 seeds). The floor then declines monotonically, $0.115 \to 0.076$, where the environment count did not, and the decline is genuine rather than a shrinking difference: MCC improves by $+0.018$ while CCA erodes only -0.010. But the effect is small (Spearman -0.13, $p = 0.25$; ill- versus well-conditioned $d = +0.33$, $p = 0.21$, against a design minimum detectable effect of $d = 0.74$), and no conditioning level rejects the null, the best conditioned reaching $p = 0.34$. So $\sigma_{\min}(L)$, not the environment count, is the axis to study, and the theorem's binary condition does have a continuous practical analogue; in this regime that analogue is bounded at $\approx 21\%$ of the full rotation signature. We report this as a measurement, not a refutation: the theorem concerns the population limit and does not promise that a finite-sample optimiser finds the identified solution.

8 Application: overlap governs recovery only through the objective

We build a generative process in which support overlap ρ and a removable batch offset δ are *provably* decoupled. Overlap is imposed symmetrically so that the pooled marginal is exactly uniform at every ρ; per-feature standardisation uses a fixed reference grid; the mixing map is non-saturating; and the observation depth is drawn independently of ρ, δ and domain. The decisive check is removability, the divergence-versus-λ split of Ben-David et al. (2010a): an affine batch map removes 100.0% of δ's expression gap and $\approx 0\%$ of ρ's.

A faithful encoder is overlap-invariant. Across a conditional VAE and an iVAE with the variability condition satisfied and violated (900 runs, 25 seeds per cell), the change in CCA $-$ MCC from $\rho = 1$ to $\rho = 0.05$ is equivalence-tested (TOST) against a pre-registered effect size and is **equivalent in all three arms** ($d = -0.23$, -0.21, $+0.15$). Stated in gap units rather than standardised units, the null bounds any ρ-dependent effect at 0.070, about a quarter of the full rotation signature. A power analysis showed the five-seed pilot could detect only $d \approx 2$, which is why the design was expanded; the analysis plan was fixed after the pilot and before the confirmatory run.

Alignment objectives collapse recovery. On the same process, a moment-matching integration penalty makes recovery overlap-dependent: CCA falls from 0.99 to 0.70, with drops of $+0.199$ ($d = 1.67$) and $+0.291$ ($d = 2.37$) at two strengths (Figure 4). The failure is stochastic, a rising *fraction* of runs folding the shifted axis as overlap shrinks. An adversarial objective reproduces it in direction and significance ($+0.101$, $d = 1.08$) but at roughly a third the magnitude, so the effect is a property of aligning rather than of one penalty, with strongly mechanism-dependent size. We report the shape conservatively: only one of four arms is strictly monotone in ρ and only one licenses a segmented-over-linear fit on BIC, so we claim a large but saturating degradation rather than a threshold, and the clean dose-response is in alignment strength rather than in overlap. Crucially, CCA $-$ MCC registers none of this (Figure 4c), which is Demonstration 2.

9 Related work

Locatello et al. (2019) established that unsupervised disentanglement is unidentifiable without inductive bias, and Khemakhem et al. (2020) gave auxiliary-variable conditions that restore identifiability; Higgins et al. (2018) framed disentanglement group-theoretically, which is the vocabulary our thesis lives in. Our contribution is neither a new metric nor a new identifiability result: it is that the invariance class of the *estimand* determines which failures are observable at all, and that this class depends on a coordinate choice. On the evaluation side, Eastwood & Williams (2018) introduced DCI and Eastwood et al. (2023) connected it to identifiability notions, and MCC is standard in nonlinear ICA. Closest in spirit is work observing that disentanglement metrics disagree with one another, but we are aware of no treatment that derives a reported

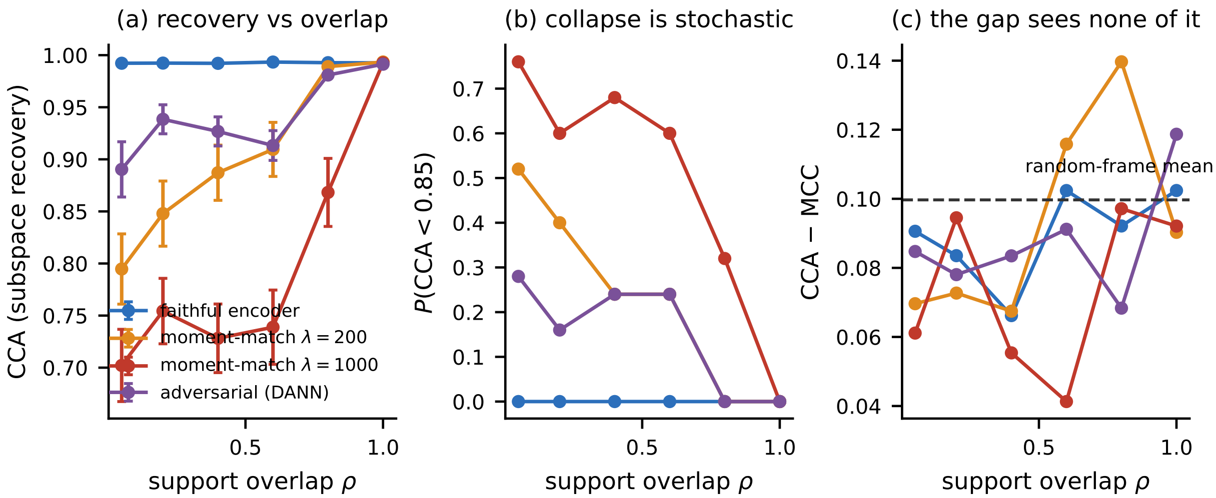

Figure 4: (a) A faithful per-sample encoder is flat in ρ; alignment objectives collapse. (b) The collapse is stochastic. (c) CCA − MCC sits on the random-frame mean throughout and is blind to the effect in (a), because the folding is approximately axis-factorised.

threshold as a metric-geometry constant, or that instruments the identifiability floor against a closed-form random-frame null. For the application, over-correction in domain integration is a recognised failure mode with dedicated methods and metrics; our point there is narrower, that whether overlap governs recovery is set by the objective. Ben-David et al. (2010a) supply the divergence/λ decomposition we use to certify that our two knobs are separable, and Ben-David et al. (2010b) the impossibility result that motivates why an unsupervised separation of the two is not available in general.

10 Discussion and limitations

The taxonomy is illustrative rather than exhaustive: three failure modes and six estimands, each row an existence proof rather than a coverage map. The geometry constant is not universal, being fixed only at a given marginal shape, embedding dimension, per-axis scale and k, and existing as a stable constant only for compact-support marginals. The quantity CCA − MCC sits on an identifiability floor rather than a zero baseline and is not interpretable per run; worse, that floor is *not subtractable*, because its height depends on the angle between the data's shift direction and the latent basis, which is exactly the unobservable quantity on real data. The recovery-invariance null is for a regime in which the shifted factor is decodable within a domain, and the paired alignment result supplies the other side. Our variability-condition measurement is bounded rather than zero, and $n_{\text{env}} = 9$ carries the most hint of an effect; resolving it would need roughly sixty to eighty seeds at the best-conditioned level. Finally, all experiments are synthetic by design, and we make no claim that these particular numbers transfer to any given real dataset.

The practical recommendation is a check with a receipt. Before reading a recovery result as a phenomenon, state the estimand's invariance class, verify that the failure mode of interest lies outside it, and for a critical-threshold claim compare the threshold against the metric's own geometry at one's own marginal, dimension, scale and k.

Reproducibility statement

All experiments are synthetic, CPU-only, and require no data download. Anonymized code and every result table are available at `https://anonymous.4open.science/r/ANON-PLACEHOLDER`, with `REPRODUCE.md` mapping each claim in this paper to the script that produces it. Because all result files are included, every analysis and figure can be regenerated in seconds without repeating the sweeps that produced them; the sweep commands are given for regeneration from scratch, and the three long campaigns are resumable and shard by seed.

Three results need no model fitting at all and run in about five minutes total: the estimand × failure-mode table of Figure 1 together with the gap = CCA − MCC identity, the oracle receipt for the threshold of §4, and the random-frame null and KS tests of §7. These carry the paper's central argument and involve no generative process or trained model, being properties of the estimands alone.

The pipeline supporting §8 is gated: each of the three verification scripts prints per-check PASS/FAIL and exits non-zero on failure, and no downstream stage was run until the stage below it passed. The retired first version of the probe is retained in the artifact alongside a document recording why each of its numbers is void, so that the retraction described in §4 is itself checkable: the oracle receipt reproduces the retracted threshold from the retired code.

Broader impact statement

This work is methodological and uses only synthetic data. Its intended effect is to make claims about latent recovery easier to check, which we expect to reduce rather than increase the risk of over-claimed representation-learning results being carried into downstream applications.

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